INCH-POUND

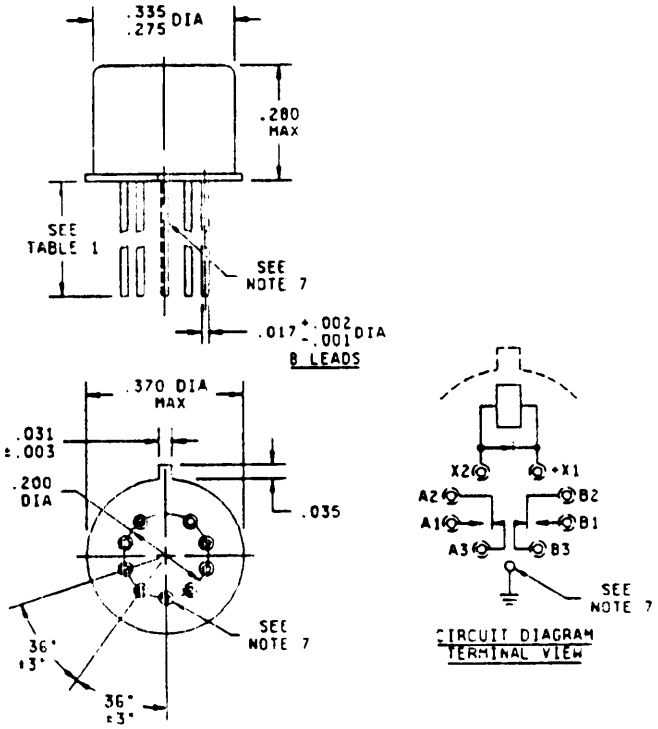
MIL-R-39016/620A
29 July 1994
SUPERSEDING
MIL-R-39016/620
24 August 1989

MILITARY SPECIFICATION SHEET

RELAYS, ELECTROMAGNETIC, ESTABLISHED RELIABILITY, DPDT, LOW LEVEL TO 1.0 AMPERE WITH INTERNAL DIODES FOR COIL TRANSIENT SUPPRESSION AND POLARITY REVERSAL PROTECTION, INCLUDING INSPECTIONS OF APPENDICES A AND B

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-R-39016.



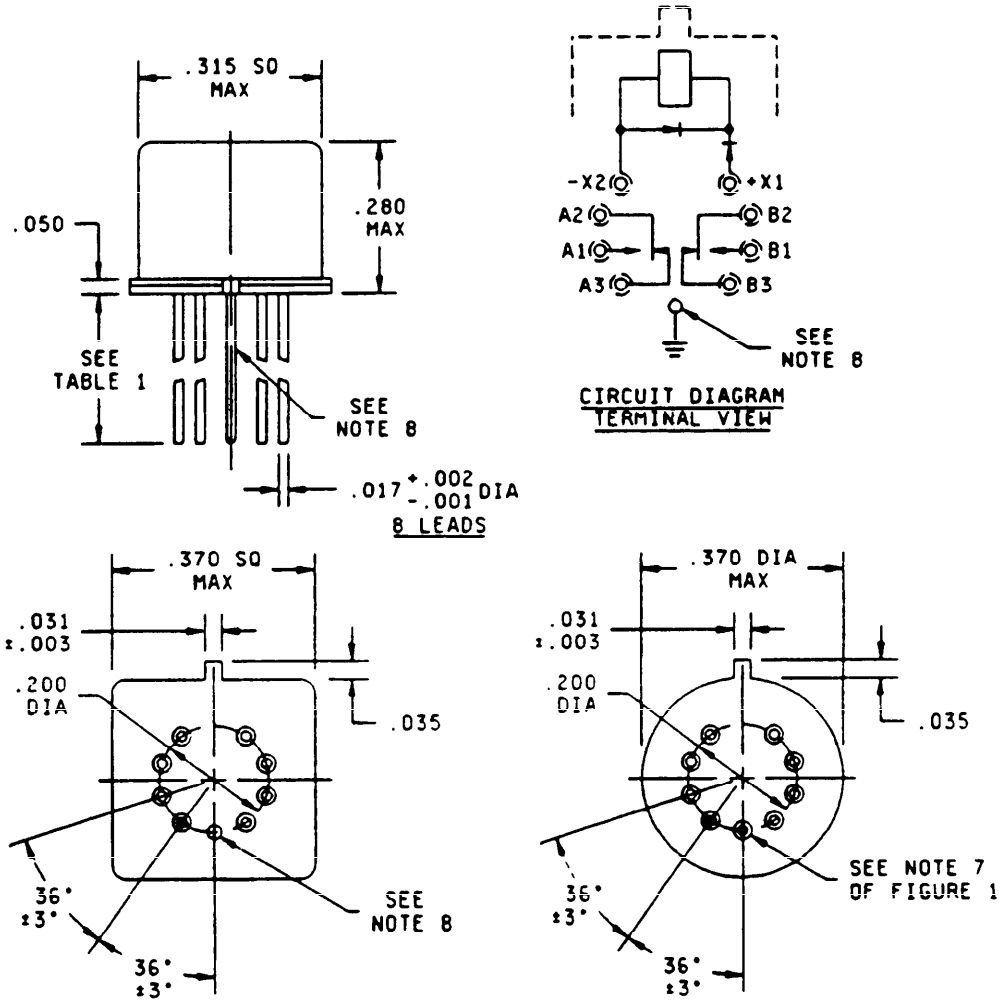
Inches	mm
.001	0.03
.002	0.05
.003	0.08
.017	0.43
.031	0.79
.035	0.89
.200	5.08
.275	6.98
.280	7.11
.335	8.51
.370	9.40

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.010 (0.25 mm).
4. Relays shall have a plus (+) sign placed on the circuit diagram as shown.
5. Coil symbol optional in accordance with MIL-STD-1285.
6. Circuit diagram shown on part is the terminal view.
7. The grounding pin shown is a noninsulated case ground (see table I for applicable dash numbers).

FIGURE 1. Dimensions and configuration (round).

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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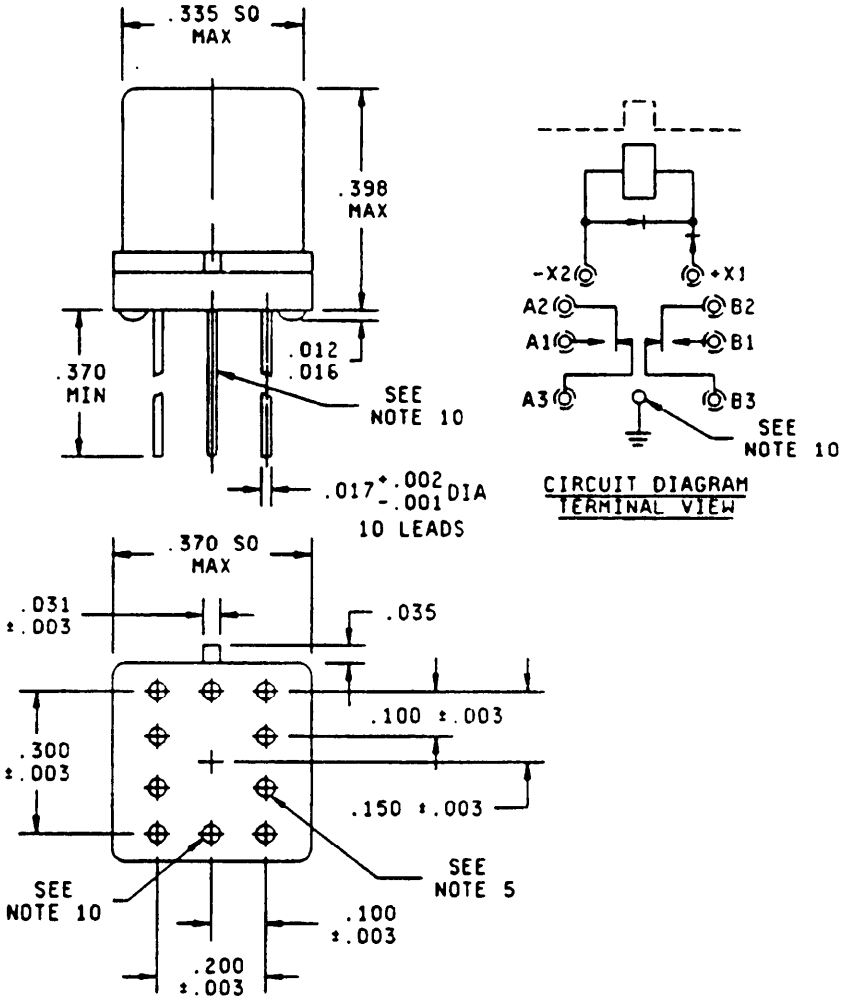


Inches	mm	Inches	mm	Inches	mm
.001	0.03	.031	0.79	.280	7.11
.002	0.05	.035	0.89	.315	8.00
.003	0.08	.050	1.27	.370	9.40
.017	0.43	.200	5.08		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.010 (0.25 mm).
4. Relays shall have a (+) sign placed on the circuit diagram as shown.
5. Coil symbol optional in accordance with MIL-STD-1285.
6. Circuit diagram shown on part is the terminal view.
7. Shape optional within the envelope dimension.
8. The grounding pin shown is a noninsulated case ground (see table I for applicable dash numbers).

FIGURE 2. Dimensions and configuration (square or round).

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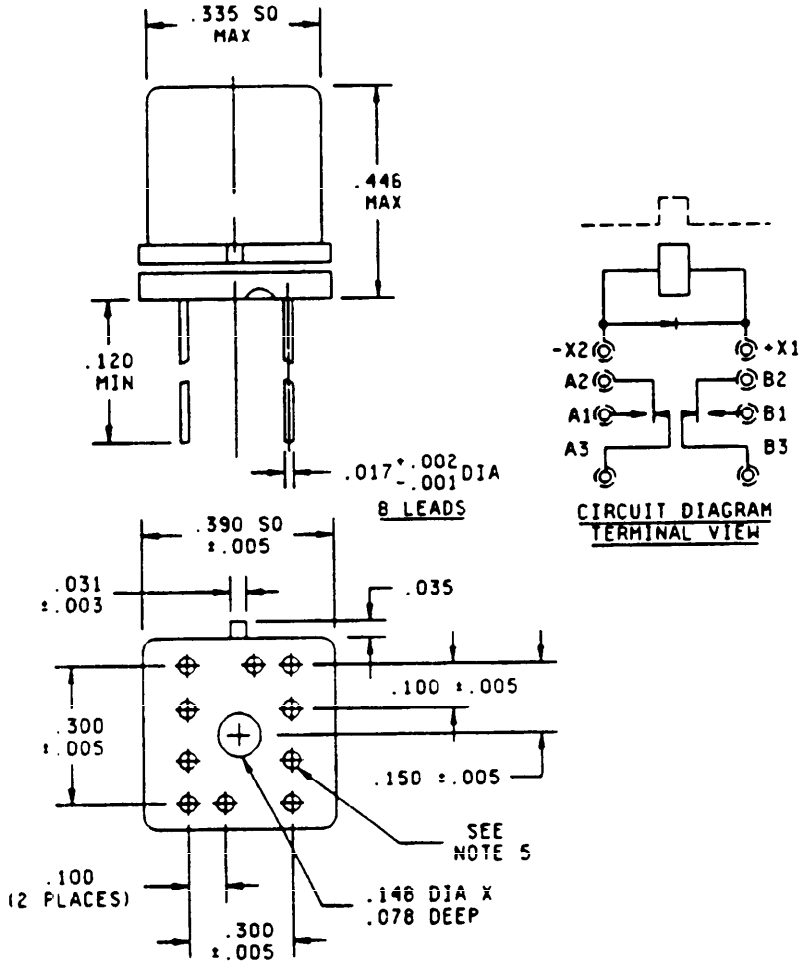


Inches	mm	Inches	mm	Inches	mm
.001	0.03	.017	0.43	.200	5.08
.002	0.05	.031	0.79	.300	7.62
.003	0.08	.035	0.89	.335	8.51
.012	0.30	.100	2.54	.370	9.40
.016	0.41	.150	3.81	.398	10.11

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.010 (0.25 mm).
4. Spreader pads shall be certified to A-A-55485/5.
5. Dimensions and tolerances shown for the bottom view of the spreader pad are for the center to center location of the holes in the spreader pad.
6. Shape optional within the envelope dimension.
7. Relays shall have a plus (+) sign placed on the circuit diagram as shown.
8. Coil symbol optional in accordance with MIL-STD-1285.
9. Circuit diagram shown on part is the terminal view.
10. The grounding pin shown is a noninsulated case ground (see table I for applicable dash numbers).

FIGURE 3. Dimensions and configuration relay with spreader pad attached.

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Inches	mm	Inches	mm	Inches	mm
.001	0.03	.031	0.79	.148	3.76
.002	0.05	.035	0.89	.300	7.62
.003	0.08	.078	1.98	.335	8.51
.005	0.13	.100	2.54	.390	9.91
.017	0.43	.120	3.05	.446	11.33

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.010 (0.25 mm).
4. Spreader pads shall be certified to A-A-55485/5.
5. Dimensions and tolerances shown for the bottom view of the spreader pad are for the center to center location of the holes in the spreader pad.
6. Shape optional within the envelope dimension.
7. Coil symbol optional in accordance with MIL-STD-1285.
8. Relays shall have a plus (+) sign placed on the circuit diagram as shown.
9. Circuit diagram shown on part is the terminal view.

FIGURE 4. Dimensions and configuration relay with spreader pad attached (square or round).

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REQUIREMENTS:

Cleaning and small particle inspection: Applicable.

CONTACT DATA:

Load ratings:

High level (relay case grounded):

Resistive:

1.0 ampere at 28 V dc.

250 mA at 115 V ac 60 Hz and 400 Hz case not grounded.

100 mA at 115 V ac 60 Hz and 400 Hz case grounded.

Inductive load: 0.2 ampere at 28 V dc with 0.32 henry inductance.

Lamp load: 0.10 ampere at 28 V dc.

Low level: 10 µA to 50 µA at 10 mV to 50 mV dc or peak ac.

Intermediate current: Applicable.

Contact resistance or voltage drop:

Initial: 0.100 ohm maximum (0.125 ohm maximum with figure 3 spreader pad attached and 0.150 ohm maximum with figure 4 spreader pad attached).

High level:

During life: Not more than 5 percent of open circuit voltage.

After life: 0.200 ohm maximum (0.225 ohm maximum with figure 3 spreader pad attached and 0.250 ohm maximum with figure 4 spreader pad attached).

Low level:

During life: 33 ohms maximum.

After life: 0.150 ohm maximum (0.175 ohm maximum with figure 3 spreader pad attached and 0.200 ohm maximum with figure 4 spreader pad attached).

Intermediate current:

During life: 1 ohm maximum.

After life: 0.200 ohm maximum (0.225 ohm maximum with figure 3 spreader pad attached and 0.250 ohm maximum with figure 4 spreader pad attached).

Contact bounce: 1.5 ms maximum (applicable after high level rated load life and intermediate current tests).

Contact stabilization time: 2.0 ms maximum.

Overload (high level only): Two times rated resistive current. Not applicable to ac load ratings.

COIL DATA: See table I.

Operate time: 2.0 ms maximum over temperature range with rated coil voltage.

Release time: 4.0 ms maximum over temperature range from rated coil voltage.

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ELECTRICAL DATA:

Insulation resistance: 10,000 megohms minimum at 500 V dc, except the resistance between coil and case at high temperature shall be 1,000 megohms minimum. 1/

Dielectric withstanding voltage: 1/

Sea level:

Between case, frame or enclosure, and all contacts; both in the energized and deenergized positions	500 V rms (60 Hz).
Between case, frame or enclosure, and coils	500 V rms (60 Hz).
Between all contacts and coils	500 V rms (60 Hz).
Between open contacts in the energized and deenergized positions	500 V rms (60 Hz).
Between open contacts	500 V rms (60 Hz).

Altitude:

Between all terminals to case	125 V rms (60 Hz).

DIODE CHARACTERISTICS: 1/

Coil transient suppression: Applicable.

Diode breakdown and block integrity (delete coil resistance and substitute this test in all inspection tables of MIL-R-39016): With applicable voltage applied to the relay coil circuit in the reverse direction, monitor leakage current with dc microammeter, oscilloscope, or qualifying activity approved test equipment. Leakage current shall not exceed the specified value.

Maximum negative transient: 1.0 volt.

Breakdown voltage: 100 V dc at 10 µA. (This test may be performed in-process or as final assembly.)

Maximum leakage current: 1 µA at 50 V dc.

Semiconductor in process screening: Applicable, visual inspection of semiconductors shall be in accordance with method 2073 or 2074, MIL-STD-750.

ENVIRONMENTAL DATA:

Temperature range: -65°C to +125°C.

Vibration (sinusoidal): Method 204, MIL-STD-202. Contact chatter shall not exceed 10 µs maximum for closed contacts and 1 µs maximum closure for open contacts.

Vibration (random): Method 214, MIL-STD-202, test condition IG. Contact chatter shall not exceed 10 µs maximum for closed contacts and 1 µs maximum closure for open contacts. Applicable to qualification, group A and group C inspection testing.

Shock (specified pulse): Method 213, MIL-STD-202, test condition B (75 g's). Contact chatter shall not exceed 10 µs maximum for closed contacts and 1 µs maximum closure for open contacts.

1/ Insulation resistance and dielectric withstanding voltage tests must always precede all other specified electrical measurements. Connect all coil terminals together to avoid damage to diodes.

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Magnetic interference: Applicable.

Resistance to soldering heat: Applicable.

Acceleration: Applicable.

Salt atmosphere (corrosion): In accordance with method 1041, MIL-STD-750.

Particle Impact Noise Detection: Applicable.

PHYSICAL DATA:

Terminal strength: Method 211, MIL-STD-202.

Pull test: Test condition A, 1 pound pull.

Bend test: Test condition C, 0.5 pound pull.

Twist test: As specified in MIL-R-39016.

Solderability: Applicable.

Dimensions and configuration: See figures 1, 2, 3, and 4.

Weight: 2.27 grams (0.08 ounce) maximum, 2.52 grams (0.089 ounce) maximum with spreader pad attached.

Seal: Hermetic, adjunct sealant shall not be used.

Minimum marking: Military Part or Identifying Number (PIN) "J" with the date code (example: J9237), circuit diagram, manufacturer's name or source code.

Outgassing: External materials (including marking ink) used in the finished relay shall be tested in accordance with ASTM-E-595. Samples to be tested shall have been processed in the same manner as that used in production of the qualification lot. Materials, when tested, shall not release greater than 1.0 percent total mass loss (TML) and 0.1 percent collected volatile condensable material (CVCM). Materials data listed in the latest revision of the NASA Reference Publication 1124 which meet these TML and CVCM requirements may be substituted in lieu of testing.

LIFE TEST REQUIREMENTS:

High level: 100,000 cycles per contact load rating per relay.

Low level: 100,000 cycles per relay.

PART OR IDENTIFYING NUMBER (PIN): M39016/620- (plus dash number from table I and suffix letter designation failure rate level).

QUALIFICATION PROCEDURE: 2/

To obtain qualification to MIL-R-39016/620, a manufacturer shall:

a. Currently possess listing on QPL-39016 for all corresponding MIL-R-39016/20 dash numbers intended to be tested to MIL-R-39016/620 requirements.

2/ MIL-STD-790 requirements are applicable. Qualification at the initial failure rate level shall be granted based on corresponding MIL-R-39016/20 qualification. Retention of qualification and extension of qualification to lower failure rate level shall be based on test data accumulated on MIL-R-39016/20.

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b. Possess qualifying activity approved Particle Impact Noise Detection (PIND) and Cleaning and small particle inspection testing capabilities. At the discretion of the qualifying activity, approval may be granted based upon the following:

(1) An audit of the manufacturer's testing capabilities by the qualifying activity.

(2) The submission, to the qualifying activity, of procedures used in the performance of the PIND and Cleaning and small particle inspection tests.

(3) The submission, to the qualifying activity, of test data in accordance with the PIND and Cleaning and small inspection testing requirements specified in MIL-R-39016.

c. Submit to the qualifying activity, certification of intent to supply product qualified to MIL-R-39016/620.

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TABLE I. Dash numbers and characteristics. 1/

Dash numbers 2/									At +25°C					Over temperature range		
Lead length .187 +.040 -.010	Lead length .500 min	Lead length .500 min with ground	Spreader pads 3/	Spreader pads with ground 3/	Figure	Coil voltage (V dc) 4/		Coil resistance (reference only) ohms	Coil circuit current (mA) 5/		Specified pickup value (voltage) (V dc)	Specified hold value (voltage) (V dc)	Specified dropout value (voltage) (V dc)	Specified pickup value (voltage) (V dc)	Specified hold value (voltage) (V dc)	Specified dropout value (voltage) (V dc)
						Rated	Max		Max	Min						
001	002	003			1	5.0	5.8	39	128.2	93.2	3.2	2.3	0.6	4.0	2.8	0.6
004	005	006				6.0	8.0	78	78.3	58.3	4.0	2.8	0.7	5.0	3.4	0.7
007	008	009				9.0	12.0	220	42.9	33.0	6.3	4.2	0.9	7.8	5.3	0.8
010	011	012				12.0	16.0	390	32.8	25.6	8.0	5.2	1.1	10.0	6.5	0.9
013	014	015				18.0	24.0	880	22.1	17.5	11.5	7.3	1.4	14.5	10.0	1.1
016	017	018				26.5	32.0	1,560	18.5	14.8	15.2	9.5	1.8	19.0	13.0	1.4
019	020	021			2	5.0	5.8	39	128.2	93.2	3.2	2.3	0.6	4.0	2.8	0.6
022	023	024				6.0	8.0	78	78.3	58.3	4.0	2.8	0.7	5.0	3.4	0.7
025	026	027				9.0	12.0	220	42.9	33.0	6.3	4.2	0.9	7.8	5.3	0.8
028	029	030				12.0	16.0	390	32.8	25.6	8.0	5.2	1.1	10.0	6.5	0.9
031	032	033				18.0	24.0	880	22.1	17.5	11.5	7.3	1.4	14.5	10.0	1.1
034	035	036				26.5	32.0	1,560	18.5	14.8	15.2	9.5	1.8	19.0	13.0	1.4
			037	038	3	5.0	5.8	39	128.2	93.2	3.2	2.3	0.6	4.0	2.8	0.6
			039	040		6.0	8.0	78	78.3	58.3	4.0	2.8	0.7	5.0	3.4	0.7
			041	042		9.0	12.0	220	42.9	33.0	6.3	4.2	0.9	7.8	5.3	0.8
			043	044		12.0	16.0	390	32.8	25.6	8.0	5.2	1.1	10.0	6.5	0.9
			045	046		18.0	24.0	880	22.1	17.5	11.5	7.3	1.4	14.5	10.0	1.1
			047	048		26.5	32.0	1,560	18.5	14.8	15.2	9.5	1.8	19.0	13.0	1.4
			049		4	5.0	5.8	39	128.2	93.2	3.2	2.3	0.6	4.0	2.8	0.6
			050			6.0	8.0	78	78.3	58.3	4.0	2.8	0.7	5.0	3.4	0.7
			051			9.0	12.0	220	42.9	33.0	6.3	4.2	0.9	7.8	5.3	0.8
			052			12.0	16.0	390	32.8	25.6	8.0	5.2	1.1	10.0	6.5	0.9
			053			18.0	24.0	880	22.1	17.5	11.5	7.3	1.4	14.5	10.0	1.1
			054			26.5	32.0	1,560	18.5	14.8	15.2	9.5	1.8	19.0	13.0	1.4

1/ Each relay possesses high level and low level capabilities. However, relays previously tested or used above 10 mA resistive at 6 V dc maximum or peak ac open circuits not recommended for subsequent use in low level applications.

2/ The suffix letter M, P, or R to designate the applicable failure rate level shall be added to the applicable listed dash number. Failure rate level (percent per 10,000 cycles): M, 1.0; P, 0.1; R, 0.01. Example: 007M - - - - 054R.

3/ Relays supplied with spreader pads (-037 through -054) shall have the pad rigidly attached.

4/ Caution: The use of any coil voltage less than the rated coil voltage will compromise the operation of the relay.

5/ Coil resistance not directly measurable at relay terminals. When rated voltage is applied to the coil terminals, the coil circuit current must be within the limits shown. Measure at 25°C at rated voltage within 5 seconds, maximum.

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CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85
NASA - NA

Review activities:
Army - AR
Navy - AS, MC, OS, SH
Air Force - 19, 99
NSA - NS

Preparing activity:
DLA - ES

(Project 5945-0918-05)

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